Advanced DBMS Assignment

by: *Lovesh Kumrawat (Roll No: DS7E-2507)*

1

```
SELECT std_name FROM student WHERE marks > 80;
```

```
mysql> SELECT std_name FROM student WHERE marks > 80;
+----------+
| std_name |
+----------+
| Vinit    |
| Raj      |
+----------+
2 rows in set (0.01 sec)

mysql> |
```

2

```
SELECT subject, COUNT(*) AS Student_Count FROM student GROUP BY subject;
```

```
mysql> SELECT subject, COUNT(*) AS Student_Count FROM student GROUP BY subject;
+---------+---------------+
| subject | Student_Count |
+---------+---------------+
| DBMS    |             3 |
| Python  |             2 |
| OR      |             1 |
+---------+---------------+
3 rows in set (0.10 sec)
```

3

```
SELECT std_name FROM student WHERE subject = 'DBMS';
```

```
mysql> SELECT std_name FROM student WHERE subject = 'DBMS';
+----------+
| std_name |
+----------+
| Amit     |
| Vinit    |
| Ajay     |
+----------+
3 rows in set (0.17 sec)
```

4

```
SELECT subject, AVG(marks) AS average_marks FROM student GROUP BY subject;
```

```
mysql> SELECT subject, AVG(marks) AS average_marks FROM student GROUP BY subject;
+---------+---------------+
| subject | average_marks |
+---------+---------------+
| DBMS    |       83.3333 |
| Python  |       80.0000 |
| OR      |       60.0000 |
+---------+---------------+
3 rows in set (0.02 sec)
```

5

```
SELECT std_name, marks FROM student ORDER BY marks DESC;
```

```
mysql> SELECT std_name, marks FROM student ORDER BY marks DESC;
+----------+-------+
| std_name | marks |
+----------+-------+
| Vinit    |    90 |
| Raj      |    90 |
| Amit     |    80 |
| Ajay     |    80 |
| Sumit    |    70 |
| Deepak   |    60 |
+----------+-------+
6 rows in set (0.02 sec)
```

6

```
SELECT std_name, marks FROM student ORDER BY marks DESC LIMIT 1;
```

```
mysql> SELECT std_name, marks FROM student ORDER BY marks DESC LIMIT 1;
+----------+-------+
| std_name | marks |
+----------+-------+
| Raj      |    90 |
+----------+-------+
1 row in set (0.00 sec)
```

7

```
SELECT std_name FROM student WHERE marks < (SELECT AVG(marks) FROM student);
```

```
mysql> SELECT std_name FROM student WHERE marks < (SELECT AVG(marks) FROM student);
+----------+
| std_name |
+----------+
| Sumit    |
| Deepak   |
+----------+
2 rows in set (0.05 sec)
```

8

```
SELECT SUM(marks) AS Total_Marks FROM student;
```

```
mysql> SELECT SUM(marks) AS Total_Marks FROM student;
+-------------+
| Total_Marks |
+-------------+
|         470 |
+-------------+
1 row in set (0.01 sec)
```

9

```
SELECT subject, MAX(marks) AS highest_score FROM student GROUP BY subject;
```

```
mysql> SELECT subject, MAX(marks) AS highest_score FROM student GROUP BY subject;
+---------+---------------+
| subject | highest_score |
+---------+---------------+
| DBMS    |            90 |
| Python  |            90 |
| OR      |            60 |
+---------+---------------+
3 rows in set (0.01 sec)
```

10

```
SELECT COUNT(*) AS students_below_50 FROM student WHERE marks < 50;
```

```
mysql> SELECT COUNT(*) AS students_below_50 FROM student WHERE marks < 50;
+-------------------+
| students_below_50 |
+-------------------+
|                 0 |
+-------------------+
1 row in set (0.06 sec)
```

11

```
SELECT subject, AVG(marks) AS average_marks FROM student GROUP BY subject;
```

```
mysql> SELECT subject, AVG(marks) AS average_marks FROM student GROUP BY subject;
+---------+---------------+
| subject | average_marks |
+---------+---------------+
| DBMS    |       83.3333 |
| Python  |       80.0000 |
| OR      |       60.0000 |
+---------+---------------+
3 rows in set (0.01 sec)
```

12

```
SELECT COUNT(DISTINCT subject) AS distinct_courses FROM student;
```

```
mysql> SELECT COUNT(DISTINCT subject) AS distinct_courses FROM student;
+------------------+
| distinct_courses |
+------------------+
|                3 |
+------------------+
1 row in set (0.02 sec)
```

13

```
SELECT subject, COUNT(*) AS student_count FROM student GROUP BY subject;
```

```
mysql> SELECT subject, COUNT(*) AS student_count FROM student GROUP BY subject;
+---------+---------------+
| subject | student_count |
+---------+---------------+
| DBMS    |             3 |
| Python  |             2 |
| OR      |             1 |
+---------+---------------+
3 rows in set (0.01 sec)
```

14

```
SELECT MIN(marks) AS minimum_marks FROM student;
```

```
mysql> SELECT MIN(marks) AS minimum_marks FROM student;
+---------------+
| minimum_marks |
+---------------+
|            60 |
+---------------+
1 row in set (0.01 sec)
```

15

```
SELECT std_name, SUM(marks) AS total_marks FROM student GROUP BY std_name;
```

```
mysql> SELECT std_name, SUM(marks) AS total_marks FROM student GROUP BY std_name;
+----------+-------------+
| std_name | total_marks |
+----------+-------------+
| Amit     |          80 |
| Sumit    |          70 |
| Vinit    |          90 |
| Deepak   |          60 |
| Ajay     |          80 |
| Raj      |          90 |
+----------+-------------+
6 rows in set (0.00 sec)
```

16

```
SELECT COUNT(*) AS students_above_average FROM student WHERE marks > (SELECT AVG(marks) FROM student);
```

```
mysql> SELECT COUNT(*) AS students_above_average FROM student WHERE marks > (SELECT AVG(marks) FROM student);
+------------------------+
| students_above_average |
+------------------------+
|                      4 |
+------------------------+
1 row in set (0.00 sec)
```

17

```
SELECT subject FROM student GROUP BY subject ORDER BY AVG(marks) DESC LIMIT 1;
```

```
mysql> SELECT subject FROM student GROUP BY subject ORDER BY AVG(marks) DESC LIMIT 1;
+---------+
| subject |
+---------+
| DBMS    |
+---------+
1 row in set (0.02 sec)
```

18

```
SELECT std_name, marks, (marks / 100.0) * 100 AS percentage FROM student;
```

```
mysql> SELECT std_name, marks, (marks / 100.0) * 100 AS percentage FROM student;
+----------+-------+------------+
| std_name | marks | percentage |
+----------+-------+------------+
| Amit     |    80 |    80.0000 |
| Sumit    |    70 |    70.0000 |
| Vinit    |    90 |    90.0000 |
| Deepak   |    60 |    60.0000 |
| Ajay     |    80 |    80.0000 |
| Raj      |    90 |    90.0000 |
+----------+-------+------------+
6 rows in set (0.01 sec)
```

19

```
SELECT std_name, ROUND(marks) AS rounded_marks FROM student;
```

```
mysql> SELECT std_name, ROUND(marks) AS rounded_marks FROM student;
+----------+---------------+
| std_name | rounded_marks |
+----------+---------------+
| Amit     |            80 |
| Sumit    |            70 |
| Vinit    |            90 |
| Deepak   |            60 |
| Ajay     |            80 |
| Raj      |            90 |
+----------+---------------+
6 rows in set (0.00 sec)
```

20

```
SELECT MAX(marks) - MIN(marks) AS Marks_Difference FROM student;
```

```
mysql> SELECT MAX(marks) - MIN(marks) AS Marks_Difference FROM student;
+------------------+
| Marks_Difference |
+------------------+
|               30 |
+------------------+
1 row in set (0.00 sec)
```

21

```
SELECT std_name, marks FROM student WHERE marks > (SELECT AVG(marks) * 1.10 FROM student);
```

```
mysql> SELECT std_name, marks FROM student WHERE marks > (SELECT AVG(marks) * 1.10 FROM student);
+----------+-------+
| std_name | marks |
+----------+-------+
| Vinit    |    90 |
| Raj      |    90 |
+----------+-------+
2 rows in set (0.00 sec)
```

22

```
SELECT std_name, marks, POWER(marks, 2) AS marks_squared FROM student;
```

```
mysql> SELECT std_name, marks, POWER(marks, 2) AS marks_squared FROM student;
+----------+-------+---------------+
| std_name | marks | marks_squared |
+----------+-------+---------------+
| Amit     |    80 |          6400 |
| Sumit    |    70 |          4900 |
| Vinit    |    90 |          8100 |
| Deepak   |    60 |          3600 |
| Ajay     |    80 |          6400 |
| Raj      |    90 |          8100 |
+----------+-------+---------------+
6 rows in set (0.05 sec)
```

23

```
SELECT ROUND(AVG(marks), 2) AS average_marks FROM student;
```

```
mysql> SELECT ROUND(AVG(marks), 2) AS average_marks FROM student;
+---------------+
| average_marks |
+---------------+
|         78.33 |
+---------------+
1 row in set (0.01 sec)
```

24

```
SELECT std_name, marks, (marks / (SELECT SUM(marks) FROM student) * 100) AS percentage_contribution FROM student;
```

```
mysql> SELECT std_name, marks, (marks / (SELECT SUM(marks) FROM student) * 100) AS percentage_contribution FROM studen
+----------+-------+-------------------------+
| std_name | marks | percentage_contribution |
+----------+-------+-------------------------+
| Amit     |    80 |                 17.0213 |
| Sumit    |    70 |                 14.8936 |
| Vinit    |    90 |                 19.1489 |
| Deepak   |    60 |                 12.7660 |
| Ajay     |    80 |                 17.0213 |
| Raj      |    90 |                 19.1489 |
+----------+-------+-------------------------+
6 rows in set (0.00 sec)
```

25

```
SELECT std_name, marks, ABS(marks - (SELECT AVG(marks) FROM student)) AS marks_difference FROM student;
```

```
mysql> SELECT std_name, marks, ABS(marks - (SELECT AVG(marks) FROM student)) AS marks_difference FROM student;
+----------+-------+------------------+
| std_name | marks | marks_difference |
+----------+-------+------------------+
| Amit     |    80 |           1.6667 |
| Sumit    |    70 |           8.3333 |
| Vinit    |    90 |          11.6667 |
| Deepak   |    60 |          18.3333 |
| Ajay     |    80 |           1.6667 |
| Raj      |    90 |          11.6667 |
+----------+-------+------------------+
6 rows in set (0.00 sec)
```

26

```
SELECT std_name, marks, RANK() OVER (ORDER BY marks DESC) AS std_rank FROM student;
```

```
mysql> SELECT std_name, marks, RANK() OVER (ORDER BY marks DESC) AS std_rank FROM student;
+----------+-------+----------+
| std_name | marks | std_rank |
+----------+-------+----------+
| Vinit    |    90 |        1 |
| Raj      |    90 |        1 |
| Amit     |    80 |        3 |
| Ajay     |    80 |        3 |
| Sumit    |    70 |        5 |
| Deepak   |    60 |        6 |
+----------+-------+----------+
6 rows in set (0.00 sec)
```

27

```
SELECT COUNT(*) AS total_students FROM student;
```

```
mysql> SELECT COUNT(*) AS total_students FROM student;
+----------------+
| total_students |
+----------------+
|              6 |
+----------------+
1 row in set (0.04 sec)
```

28

```
SELECT std_name, LENGTH(std_name) AS first_name_length FROM student;
```

```
mysql> SELECT std_name, LENGTH(std_name) AS first_name_length FROM student;
+----------+-------------------+
| std_name | first_name_length |
+----------+-------------------+
| Amit     |                 4 |
| Sumit    |                 5 |
| Vinit    |                 5 |
| Deepak   |                 6 |
| Ajay     |                 4 |
| Raj      |                 3 |
+----------+-------------------+
6 rows in set (0.00 sec)
```

29

```
SELECT UPPER(std_name) AS uppercase_name FROM student;
```

```
mysql> SELECT UPPER(std_name) AS uppercase_name FROM student;
+----------------+
| uppercase_name |
+----------------+
| AMIT           |
| SUMIT          |
| VINIT          |
| DEEPAK         |
| AJAY           |
| RAJ            |
+----------------+
6 rows in set (0.01 sec)
```

30

```
SELECT LOWER(std_name) AS lowercase_student_name FROM student;
```

```
mysql> SELECT LOWER(std_name) AS lowercase_student_name FROM student;
+------------------------+
| lowercase_student_name |
+------------------------+
| amit                   |
| sumit                  |
| vinit                  |
| deepak                 |
| ajay                   |
| raj                    |
+------------------------+
6 rows in set (0.01 sec)
```

31

```
SELECT std_name, SUBSTRING(std_name, 1, 2) AS first_two_letters FROM student;
```

```
mysql> SELECT std_name, SUBSTRING(std_name, 1, 2) AS first_two_letters FROM student;
+----------+-------------------+
| std_name | first_two_letters |
+----------+-------------------+
| Amit     | Am                |
| Sumit    | Su                |
| Vinit    | Vi                |
| Deepak   | De                |
| Ajay     | Aj                |
| Raj      | Ra                |
+----------+-------------------+
6 rows in set (0.00 sec)
```

32

```
SELECT * FROM student WHERE std_name LIKE 'S%';
```

```
mysql> SELECT * FROM student WHERE std_name LIKE 'S%';
+---------------+-------------+----------+---------+---------+-------+
| std_enroll_no | std_roll_no | std_name | address | subject | marks |
+---------------+-------------+----------+---------+---------+-------+
| DS1002        |        1002 | Sumit    | Bhopal  | Python  |    70 |
+---------------+-------------+----------+---------+---------+-------+
1 row in set (0.02 sec)
```

33

```
SELECT * FROM student WHERE LENGTH(std_name) > 3;
```

```
mysql> SELECT * FROM student WHERE LENGTH(std_name) > 3;
+---------------+-------------+----------+---------+---------+-------+
| std_enroll_no | std_roll_no | std_name | address | subject | marks |
+---------------+-------------+----------+---------+---------+-------+
| DS1001        |        1001 | Amit     | Indore  | DBMS    |    80 |
| DS1002        |        1002 | Sumit    | Bhopal  | Python  |    70 |
| DS1003        |        1003 | Vinit    | Dhar    | DBMS    |    90 |
| DS1004        |        1004 | Deepak   | Ujjan   | OR      |    60 |
| DS1005        |        1005 | Ajay     | Indore  | DBMS    |    80 |
+---------------+-------------+----------+---------+---------+-------+
5 rows in set (0.00 sec)
```

34

```
SELECT std_name, REPLACE(std_name, 'o', '0') AS modified_last_name FROM student;
```

```
mysql> SELECT std_name, REPLACE(std_name, 'o', '0') AS modified_last_name FROM student;
+----------+--------------------+
| std_name | modified_last_name |
+----------+--------------------+
| Amit     | Amit               |
| Sumit    | Sumit              |
| Vinit    | Vinit              |
| Deepak   | Deepak             |
| Ajay     | Ajay               |
| Raj      | Raj                |
+----------+--------------------+
6 rows in set (0.01 sec)
```

35

```
ALTER TABLE student ADD email VARCHAR(100);
```

```
mysql> ALTER TABLE student ADD email VARCHAR(100);
Query OK, 0 rows affected (0.23 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

36

```
ALTER TABLE student RENAME COLUMN std_name TO std_first_name;
```

```
mysql> ALTER TABLE student RENAME COLUMN std_name TO std_first_name;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

37

```
ALTER TABLE student DROP COLUMN subject;
```

```
mysql> ALTER TABLE student DROP COLUMN subject;
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

38

```
ALTER TABLE student ADD PRIMARY KEY (std_enroll_no);
```

```
mysql> ALTER TABLE student ADD PRIMARY KEY (std_enroll_no);
Query OK, 0 rows affected (0.19 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

39

```
ALTER TABLE student ADD CONSTRAINT unique_email UNIQUE (email);
```

```
mysql> ALTER TABLE student ADD CONSTRAINT unique_email UNIQUE (email);
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

40

```
UPDATE student SET marks = marks + 5;
```

```
mysql> UPDATE student SET marks = marks + 5;
Query OK, 6 rows affected (0.02 sec)
Rows matched: 6  Changed: 6  Warnings: 0
```

41

```
UPDATE student SET std_first_name = 'Amit Kumar' WHERE std_enroll_no = 'DS1001';
```

```
mysql> UPDATE student SET std_first_name = 'Amit Kumar' WHERE std_enroll_no = 'DS1001';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

42

```
UPDATE student SET marks = 80 WHERE marks IS NULL;
```

```
mysql> UPDATE student SET marks = 80 WHERE marks IS NULL;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

43

```
DELETE FROM student WHERE std_first_name = 'Amit Kumar';
```

```
mysql> DELETE FROM student WHERE std_first_name = 'Amit Kumar';
Query OK, 1 row affected (0.02 sec)
```

44

```
DELETE FROM student WHERE std_first_name LIKE 'A%';
```

```
mysql> DELETE FROM student WHERE std_first_name LIKE 'A%';
Query OK, 1 row affected (0.01 sec)
```

45

```
DELETE FROM student WHERE marks < 75;
```

```
mysql> DELETE FROM student WHERE marks < 75;
Query OK, 1 row affected (0.01 sec)
```

46

```
DELETE FROM student;
```

```
mysql> DELETE FROM student;
Query OK, 3 rows affected (0.02 sec)
```

47

```
SELECT * FROM student;
```

```
mysql> SELECT * FROM student;
Empty set (0.00 sec)
```

48

```
SELECT first_name, last_name FROM student;
```

```
mysql> SELECT first_name, last_name FROM student;
ERROR 1054 (42S22): Unknown column 'first_name' in 'field list'
```

49

```
SELECT * FROM student WHERE marks = 90;
```

```
mysql> SELECT * FROM student WHERE marks = 90;
+--------------+------------+----------------+---------+-------+-------+---------+
| std_enroll_no | std_roll_no | std_first_name | address | marks | email | subject |
+--------------+------------+----------------+---------+-------+-------+---------+
| DS1003       |       1003 | Vinit          | Dhar    |    90 | NULL  | DBMS    |
| DS1006       |       1006 | Raj            | Indore  |    90 | NULL  | Python  |
+--------------+------------+----------------+---------+-------+-------+---------+
2 rows in set (0.00 sec)
```

50

```
SELECT * FROM student WHERE marks > 80;
```

```
mysql> SELECT * FROM student WHERE marks > 80;
+--------------+------------+----------------+---------+-------+-------+---------+
| std_enroll_no | std_roll_no | std_first_name | address | marks | email | subject |
+--------------+------------+----------------+---------+-------+-------+---------+
| DS1003       |       1003 | Vinit          | Dhar    |    90 | NULL  | DBMS    |
| DS1006       |       1006 | Raj            | Indore  |    90 | NULL  | Python  |
+--------------+------------+----------------+---------+-------+-------+---------+
2 rows in set (0.00 sec)
```

51

```
SELECT * FROM student WHERE std_first_name = 'Raj';
```

```
mysql> SELECT * FROM student WHERE std_first_name = 'Raj';
+---------------+-------------+----------------+---------+-------+-------+---------+
| std_enroll_no | std_roll_no | std_first_name | address | marks | email | subject |
+---------------+-------------+----------------+---------+-------+-------+---------+
| DS1006        |        1006 | Raj            | Indore  |    90 | NULL  | Python  |
+---------------+-------------+----------------+---------+-------+-------+---------+
1 row in set (0.00 sec)
```

52

```
SELECT * FROM student ORDER BY marks DESC;
```

```
mysql> SELECT * FROM student ORDER BY marks DESC;
+---------------+-------------+----------------+---------+-------+-------+---------+
| std_enroll_no | std_roll_no | std_first_name | address | marks | email | subject |
+---------------+-------------+----------------+---------+-------+-------+---------+
| DS1003        |        1003 | Vinit          | Dhar    |    90 | NULL  | DBMS    |
| DS1006        |        1006 | Raj            | Indore  |    90 | NULL  | Python  |
| DS1001        |        1001 | Amit           | Indore  |    80 | NULL  | DBMS    |
| DS1005        |        1005 | Ajay           | Indore  |    80 | NULL  | DBMS    |
| DS1002        |        1002 | Sumit          | Bhopal  |    70 | NULL  | Python  |
| DS1004        |        1004 | Deepak         | Ujjan   |    60 | NULL  | OR      |
+---------------+-------------+----------------+---------+-------+-------+---------+
6 rows in set (0.01 sec)
```

53

```
SELECT COUNT(*) AS total_students FROM student;
```

```
mysql> SELECT COUNT(*) AS total_students FROM student;
+----------------+
| total_students |
+----------------+
|              6 |
+----------------+
1 row in set (0.00 sec)
```

54

```
SELECT marks, COUNT(*) AS Total_Student FROM student GROUP BY marks;
```

```
mysql> SELECT marks, COUNT(*) AS Total_Student FROM student GROUP BY marks;
+-------+---------------+
| marks | Total_Student |
+-------+---------------+
|    80 |             2 |
|    70 |             1 |
|    90 |             2 |
|    60 |             1 |
+-------+---------------+
4 rows in set (0.01 sec)
```

55

```
SELECT DISTINCT marks FROM student;
```

```
mysql> SELECT DISTINCT marks FROM student;
+-------+
| marks |
+-------+
|    80 |
|    70 |
|    90 |
|    60 |
+-------+
4 rows in set (0.01 sec)
```

56

```
SELECT * FROM student WHERE std_first_name LIKE 'A%';
```

```
mysql> SELECT * FROM student WHERE std_first_name LIKE 'A%';
+--------------+------------+----------------+---------+-------+-------+---------+
| std_enroll_no | std_roll_no | std_first_name | address | marks | email | subject |
+--------------+------------+----------------+---------+-------+-------+---------+
| DS1001       |       1001 | Amit           | Indore  |    80 | NULL  | DBMS    |
| DS1005       |       1005 | Ajay           | Indore  |    80 | NULL  | DBMS    |
+--------------+------------+----------------+---------+-------+-------+---------+
2 rows in set (0.00 sec)
```

57

```
SELECT * FROM student WHERE marks > (SELECT AVG(marks) FROM student);
```

```
mysql> SELECT * FROM student WHERE marks > (SELECT AVG(marks) FROM student);
+--------------+------------+----------------+---------+-------+-------+---------+
| std_enroll_no | std_roll_no | std_first_name | address | marks | email | subject |
+--------------+------------+----------------+---------+-------+-------+---------+
| DS1001       |       1001 | Amit           | Indore  |    80 | NULL  | DBMS    |
| DS1003       |       1003 | Vinit          | Dhar    |    90 | NULL  | DBMS    |
| DS1005       |       1005 | Ajay           | Indore  |    80 | NULL  | DBMS    |
| DS1006       |       1006 | Raj            | Indore  |    90 | NULL  | Python  |
+--------------+------------+----------------+---------+-------+-------+---------+
4 rows in set (0.01 sec)
```